

## LOUISVILLE INTL, KY, USA

WMO: 724230

Lat: 38.181N

Lon: 85.739W

Elev: 149

StdP: 99.55

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93821

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.1      | -9.1 | -19.0                         | 0.7 | -10.3 | -16.2 | 0.9 | -7.6 | 11.4                  | 5.6  | 10.4 | 5.7  | 3.7                   | 300  | 0.466 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.3                    | 34.4            | 24.0 | 33.2 | 23.8 | 32.0 | 23.3 | 25.9                | 31.7 | 25.2 | 30.9 | 24.6 | 29.9 | 4.0                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.2                            | 19.5 | 29.4 | 23.5 | 18.6 | 28.5 | 22.9 | 17.9 | 27.8 | 80.8          | 31.7 | 77.6 | 30.8 | 75.0 | 30.1 | 29.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.6           | 8.4  | 7.5 | -15.6                      | 36.3 | 4.2                | 2.0 | -18.6                                              | 37.7 | -21.0      | 38.9 | -23.4      | 40.0 | -26.4      | 41.5 |    |    |
| (5)               |      |     | -16.3                      | 27.1 | 4.0                | 0.8 | -19.2                                              | 27.7 | -21.5      | 28.2 | -23.7      | 28.6 | -26.6      | 29.3 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 14.8   | 1.6   | 4.0   | 8.8   | 14.9  | 19.9  | 24.5  | 26.2  | 25.9  | 22.1  | 15.6  | 9.0   | 4.1  | (6)  |
| (7)  |                                                                                  | DBStd       | 9.95   | 6.68  | 6.29  | 5.94  | 5.11  | 4.30  | 3.08  | 2.60  | 2.60  | 3.95  | 4.84  | 5.25  | 5.82 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 853    | 271   | 186   | 94    | 14    | 1     | 0     | 0     | 0     | 0     | 8     | 80    | 198  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 2228   | 517   | 403   | 299   | 124   | 35    | 2     | 0     | 0     | 12    | 113   | 282   | 441  | (9)  |
| (10) |                                                                                  | CDD10.0     | 2599   | 12    | 17    | 58    | 162   | 309   | 434   | 502   | 493   | 363   | 182   | 51    | 16   | (10) |
| (11) |                                                                                  | CDD18.3     | 932    | 0     | 1     | 4     | 23    | 85    | 186   | 244   | 234   | 125   | 28    | 2     | 1    | (11) |
| (12) |                                                                                  | CDH23.3     | 8914   | 0     | 2     | 33    | 185   | 695   | 1759  | 2528  | 2330  | 1143  | 228   | 10    | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 3418   | 0     | 0     | 3     | 32    | 199   | 674   | 1062  | 959   | 434   | 56    | 1     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.4    | 4.0   | 4.0   | 4.0   | 4.0   | 3.5   | 3.2   | 2.9   | 2.7   | 2.7   | 3.1   | 3.5   | 3.6  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1168   | 79    | 83    | 117   | 112   | 123   | 99    | 108   | 90    | 88    | 79    | 91    | 99   | (15) |
| (16) |                                                                                  | PrecMax     | 1752   | 184   | 268   | 379   | 355   | 294   | 257   | 255   | 223   | 277   | 253   | 194   | 225  | (16) |
| (17) |                                                                                  | PrecMin     | 829    | 11    | 19    | 26    | 19    | 35    | 12    | 9     | 16    | 2     | 10    | 18    | 16   | (17) |
| (18) |                                                                                  | PrecStd     | 205    | 40    | 50    | 66    | 66    | 58    | 54    | 54    | 48    | 60    | 53    | 43    | 46   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 19.7  | 22.2  | 26.7  | 29.4  | 31.8  | 34.2  | 36.3  | 36.2  | 35.4  | 30.7  | 24.7  | 20.8 | (19) |
| (20) |                                                                                  |             | MCWB   | 16.0  | 15.8  | 17.4  | 18.5  | 22.4  | 24.1  | 25.2  | 23.9  | 21.9  | 21.0  | 17.5  | 16.5 | (20) |
| (21) |                                                                                  | 2%          | DB     | 17.2  | 19.1  | 23.8  | 27.5  | 30.3  | 32.9  | 34.3  | 34.2  | 33.0  | 28.1  | 22.0  | 17.8 | (21) |
| (22) |                                                                                  |             | MCWB   | 14.2  | 13.9  | 15.7  | 18.0  | 21.2  | 23.9  | 24.8  | 23.7  | 22.0  | 19.8  | 16.0  | 14.4 | (22) |
| (23) |                                                                                  | 5%          | DB     | 14.8  | 16.7  | 21.0  | 25.6  | 29.0  | 31.8  | 33.0  | 32.8  | 31.2  | 25.9  | 19.7  | 15.2 | (23) |
| (24) |                                                                                  |             | MCWB   | 12.3  | 12.4  | 14.3  | 16.9  | 20.7  | 23.2  | 24.4  | 23.6  | 21.7  | 18.6  | 14.8  | 12.2 | (24) |
| (25) |                                                                                  | 10%         | DB     | 12.0  | 13.9  | 18.5  | 23.6  | 27.5  | 30.5  | 31.6  | 31.4  | 29.2  | 23.7  | 17.5  | 12.9 | (25) |
| (26) |                                                                                  |             | MCWB   | 9.0   | 9.7   | 12.8  | 16.1  | 20.3  | 22.7  | 23.8  | 23.1  | 20.9  | 17.5  | 13.3  | 10.2 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 16.8  | 17.1  | 18.3  | 21.0  | 23.9  | 26.0  | 27.4  | 26.8  | 24.5  | 23.1  | 19.0  | 17.2 | (27) |
| (28) |                                                                                  |             | MCD B  | 18.9  | 21.1  | 24.3  | 26.9  | 29.3  | 31.7  | 32.9  | 32.5  | 30.6  | 29.0  | 22.6  | 20.0 | (28) |
| (29) |                                                                                  | 2%          | WB     | 14.9  | 15.3  | 16.8  | 19.6  | 22.8  | 25.2  | 26.2  | 25.7  | 23.7  | 21.6  | 17.5  | 15.1 | (29) |
| (30) |                                                                                  |             | MCD B  | 16.8  | 18.3  | 21.9  | 24.9  | 28.3  | 31.1  | 32.2  | 31.3  | 29.6  | 25.8  | 20.7  | 17.4 | (30) |
| (31) |                                                                                  | 5%          | WB     | 12.7  | 12.9  | 15.4  | 18.4  | 21.9  | 24.5  | 25.4  | 24.9  | 23.0  | 20.0  | 15.6  | 12.8 | (31) |
| (32) |                                                                                  |             | MCD B  | 14.6  | 16.0  | 19.9  | 23.0  | 27.3  | 30.0  | 31.3  | 30.4  | 28.4  | 24.4  | 18.9  | 14.8 | (32) |
| (33) |                                                                                  | 10%         | WB     | 9.6   | 10.1  | 13.7  | 17.2  | 21.1  | 23.8  | 24.8  | 24.2  | 22.2  | 18.5  | 13.7  | 10.4 | (33) |
| (34) |                                                                                  |             | MCD B  | 12.0  | 13.3  | 17.9  | 21.9  | 25.9  | 28.9  | 30.1  | 29.2  | 27.2  | 22.6  | 16.9  | 12.5 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.3   | 9.1   | 10.1  | 11.0  | 9.9   | 9.5   | 9.3   | 9.7   | 10.6  | 10.6  | 9.5   | 7.8  | (35) |
| (36) |                                                                                  |             | MCDBR  | 10.5  | 12.3  | 13.4  | 12.9  | 11.2  | 10.6  | 10.6  | 11.2  | 12.4  | 12.6  | 12.2  | 10.5 | (36) |
| (37) |                                                                                  |             | MCWBR  | 8.5   | 9.0   | 7.6   | 6.5   | 4.8   | 4.3   | 3.8   | 3.9   | 4.4   | 6.0   | 7.8   | 8.3  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 9.8   | 11.2  | 11.7  | 10.8  | 9.6   | 9.5   | 9.5   | 9.3   | 9.9   | 10.2  | 10.5  | 9.4  | (38) |
| (39) |                                                                                  |             | MCWBR  | 8.5   | 9.4   | 7.7   | 6.6   | 4.9   | 4.4   | 4.1   | 3.9   | 4.3   | 5.7   | 8.1   | 8.4  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.315  | 0.330 | 0.360 | 0.405 | 0.445 | 0.458 | 0.482 | 0.479 | 0.423 | 0.364 | 0.331 | 0.309 | (40) |      |
| (41) |                                                                                  | taud        | 2.458  | 2.390 | 2.380 | 2.255 | 2.184 | 2.184 | 2.126 | 2.122 | 2.255 | 2.425 | 2.463 | 2.482 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 865    | 896   | 900   | 874   | 842   | 828   | 804   | 796   | 822   | 846   | 837   | 846   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 82     | 100   | 111   | 134   | 146   | 146   | 154   | 151   | 125   | 95    | 81    | 75    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.92   | 2.62  | 3.70  | 4.91  | 5.48  | 6.22  | 5.91  | 5.52  | 4.55  | 3.38  | 2.31  | 1.65  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.20   | 0.36  | 0.35  | 0.34  | 0.49  | 0.35  | 0.36  | 0.32  | 0.42  | 0.36  | 0.26  | 0.16  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.53 | N/A     | N/A    | N/A     | N/A   | -0.38 | N/A         | N/A     | +156    | +112    |
| (47) Regional (30 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +118    | +77     |

Nomenclature: See separate page